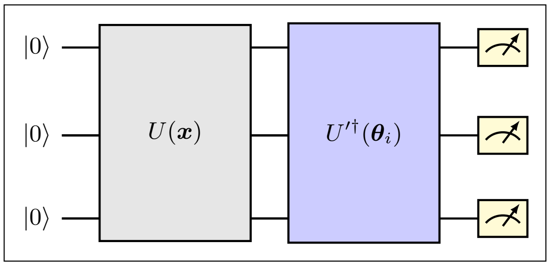

FIG. 3. Quantum circuit to estimate the transition probability between an unknown quantum state and the quantum state corresponding to a neuron. The goal of this architecture is to calculate the degree of overlap with each neuron's weights and identify the one that generates the highest number of '0 . . . 0' bitstrings.

different quantum feature maps; thus, $U = U'$ is merely an exception.

In a manner analogous to that of the kernelized self-organized map, the rule that governs the weights (i.e. angles in the parametrized quantum gates) update process is given as

$$\boldsymbol{\theta}_i^{t+1} = \boldsymbol{\theta}_i^t - \alpha^t h^t \Big(\frac{\partial}{\partial \boldsymbol{\theta}} K(\boldsymbol{\theta}, \boldsymbol{\theta}) \Big|_{\boldsymbol{\theta}_i^t} - 2 \frac{\partial}{\partial \boldsymbol{\theta}} K(\boldsymbol{x}, \boldsymbol{\theta}) \Big|_{\boldsymbol{\theta}_i^t} \Big). \quad (15)$$

In those cases where $U = U'$, the first term in Eq. (15) vanishes, since $K(\boldsymbol{\theta}, \boldsymbol{\theta}) = 1$, and thus the formula is modified to:

$$\boldsymbol{\theta}_i^{t+1} = \boldsymbol{\theta}_i^t + 2\alpha^t h^t \frac{\partial}{\partial \boldsymbol{\theta}} K(\boldsymbol{x}, \boldsymbol{\theta}) \Big|_{\boldsymbol{\theta}_i^t}. \quad (16)$$

Similar to the original version of SOM, along with the learning rate α, the neighborhood function is a hyper-parameter in the training process; however, in our case, Gaussian functional form is chosen, as given in Eq. (4). The training process results in a map $\kappa(\rho)$ from the high-dimensional space $\mathcal{B}(\mathcal{H})$ to L, which preserves the topological structure present in $\mathcal{B}(\mathcal{H})$. In the inference phase, the weights are no longer updated, but simply the appropriate BMU l^* for a data sample $\rho(\boldsymbol{x})$ is selected. The gradients of the kernel functions, as specified in Eq. (16) can be calculated using the parameter shift rule [53] given by

$$\frac{\partial}{\partial \theta} K(x, \theta) \Big|_{\theta_i^t} = \frac{K(x, \theta + \phi) - K(x, \theta - \phi)}{2 \sin(\omega\phi)/\omega}, \quad (17)$$

where ω corresponds to the eigenvalues of the Pauli gates.

IV. NUMERICAL EXPERIMENTS

In order to establish the efficacy of the proposed variational quantum self-organizing map, we employ this algorithm to generate two distinct low-dimensional topology-preserving maps. One map is constructed using classical data, while the other map is constructed using quantum data. In the first scenario, the lower-dimensional map is employed for the purpose of clustering and for creating a low-dimensional topology-preserving representation of the three distinct species of flowers from the Iris data set [54]. In the second scenario, the proposed algorithm is employed to generate a low-dimensional projection map of the state space of a 1 + 1-dimensional lattice gauge theory describing the interaction between electrons and photons, called Schwinger model [55]. The resultant low-dimensional map is employed to distinguish between two distinct phases of the model.

A. Variational QSOM on Iris data

The Iris dataset, developed by Ronald Fisher, is widely recognized and frequently employed within the machine learning field. The dataset comprises of 150 samples and 4 features. The dataset contains 50 samples from each of the three Iris species, namely Iris setosa, Iris virginica, and Iris versicolor. The four features provide data on the measurements of the sepals and petals, specifically their length and width, for each sample. The dataset also contains an appropriate label which corresponds to a particular type of the species of the Iris flower. However, note that, we do not use this label during the training phase. Our primary aim, in fact, is to evaluate the unsupervised learning capability of the algorithm which utilizes only the raw features of the data. In the training phase the algorithm is trained to generate hidden clusters in the data set, and in the inference phase a test sample is assigned to an appropriate cluster. The resultant cluster label is then compared against the corresponding label for that sample.

For the data encoding of the classical data vector and the weight vector into the quantum states, we utilize *ZZFeatureMap* available in Qiskit [56]. We use the nearest neighbour (i.e. 'pairwise') entanglement. The corresponding unitary is given as

$$U_{\Phi(\mathbf{x})} = U(\mathbf{x}) H^{\otimes n}, \quad (18)$$

where

$$U(\mathbf{x}) = \exp\Big(i \sum_i \phi(x_i) Z_i + i \sum_{<ij>} \phi'(x_i, x_j) Z_i Z_j \Big). \quad (19)$$

In principle, it is possible to customize the functions ϕ and ϕ'; however, in our experiments, we use $\phi(x_i) = 2x_i$ and $\phi'(x_i, x_{i+1}) = 2(x_i - \pi)(x_{i+1} - \pi)$. The input data is scaled from -1 to 1. It is conjectured that for such a map, the transition probability $|\langle \Psi(\boldsymbol{\theta}_i) | \Psi(\boldsymbol{x}) \rangle|^2$ (i.e. the overlap between the input quantum state and the quantum state parametrized by the weight vector), which can be estimated on a quantum computer relatively easily, is hard to evaluate on a classical computer [57].

The output layer in our architecture consists of 36 neurons arranged in a grid of size $(6, 6)$. The weight vectors